



Eliciting expert judgements to estimate risk and protective factors for Piscirickettsiosis in Chilean salmon farming

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ABSTRACT

Global production of farmed salmon is increasingly threatened by emerging infectious diseases. Piscirickettsiosis or Salmonid Rickettsial Septicemia (SRS), a disease caused by the bacterium *Piscirickettsia salmonis*, is the major responsible for the overall disease-specific mortalities in the Chilean salmon industry. In this study, we applied a structured expert elicitation process to identify risk and protective factors associated with severe outbreaks of SRS during a production cycle. We used a qualitative based-expert approach to calculate risk estimators for ten risk factors and seven protective factors. In the expert elicitation process, each participant independently estimated factors in two rounds. Between the first and second round, we facilitated a workshop among experts to discuss preliminary results. As a result, the inter-expert variation of the experts' estimates was systematically reduced. Our results are concordant with preliminary studies about risk factors for SRS. Importantly, we identified novel factors that may be associated to an increased risk for severe SRS outbreaks, such as fouling of cages, density of farms in neighborhoods, presence of sea lions and early mature salmonids. Novel factors that appear to reduce the risk of severe outbreaks of SRS were synchronized farm rest periods, opportune diagnosis and necropsy training. Using these results, we propose an intervention model to provide better information for strategic decision making.

1. Introduction

Aquaculture of farmed salmon is one of the fastest growing food production systems in the world, providing an efficient source of animal protein (Shepherd and Little, 2014). However, the global salmon sector is increasingly threatened by emerging infectious diseases, which have caused substantial environmental problems and costs for the industry (Cid Aguayo and Barriga, 2016; Pettersen et al., 2015). In Chile, the second largest salmon producer in the world, Piscirickettsiosis or

Salmonid Rickettsial Septicemia (SRS) is currently the major disease threatening the salmon industry, being responsible for extensive economic losses (Rozas and Enríquez, 2014). SRS has caused most of the total disease-specific salmon mortality between 2010 and 2016, particularly in Atlantic salmon (*Salmo salar*) and rainbow trout (*Oncorhynchus mykiss*) (Mardones et al., 2018). Therefore, identifying risk and protective factors associated to severe outbreaks of SRS is a critical challenge for sanitary management in salmon aquaculture (Gustafson et al., 2013; Mardones et al., 2018).

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SRS is mostly reported in salt and estuarine waters, being considered very rare in fresh water farms (Lannan and Fryer, 1994). The bacterium spreads primarily through horizontal transmission, mostly via the gills, followed by the skin, and to a lesser extent by the intestine (Almendras and Fuentealba, 1997; Rozas-Serri et al., 2017; Smith et al., 2004). Vaccines against SRS have delayed the appearance severe outbreaks, but do not prevent them (Jakob et al., 2014; Marshall and Tobar, 2014). Consequently, control of SRS is based heavily on antimicrobials, being responsible for 92% of 382 tons of antibiotics administered to farmed salmon in the Chilean industry by 2016 (Henríquez et al., 2016; SERNAPESCA, 2017).

A number of factors influence the manifestation of clinical disease associated with SRS, including environmental stressors (i.e., storms, algal blooms, predator attacks, low oxygen, and fluctuations in water temperatures) (Branson and Nieto Díaz-Muñoz, 1991; Cusack et al., 2002; Larenas et al., 1995; Rees et al., 2014), co-infection with other pathogens (Gaggero et al., 1995), skin damage and genetics (Lhorente et al., 2014; Yáñez et al., 2013), husbandry-related factors (Rozas and Enríquez, 2014), and a 'neighbor effect' (Rees et al., 2014). Despite these advances on studying specific drivers of infection, there is no systematic analysis of risk and protective factors associated to severe outbreaks of SRS (Mardones et al., 2018). This gap may be partly due to the lack of quantitative information available for some hypothetically relevant factors, such as stress associated to fouling of cage or presence of sea lions.

Expert elicitation techniques have been increasingly applied to inform decision making when data are insufficient (Burgman, 2005; Mellers et al., 2014), providing a useful tool to identify, estimate and rank drivers of change and risks (Sutherland and Burgman, 2015). Examples include applications in ecology and conservation (Drescher et al., 2013; Irvine et al., 2009; Kuhnert et al., 2010; Low Choy et al., 2009; Martin et al., 2012), agriculture and biogeochemistry (Krueger et al., 2012), threatened species assessment and population trends (Adams-Hosking et al., 2016; Dulvy et al., 2014; McBride et al., 2012; Short et al., 2011), and infectious diseases in aquaculture (Gustafson et al., 2005, 2010, 2013, 2014; Martin et al., 2007).

Unfortunately, unstructured expert elicitation processes are common in environmental applications (Krueger et al., 2012; Regan et al., 2005). Expert judgments that are not based on robust and proven methods are subject to individual and group heuristic bias, particularly overconfidence, group status and anchor effects (Burgman, 2004, 2016; Hemming et al., 2018a; Slovic, 1999).

In a structured expert elicitation process, based on validated methodologies, each step is considered as a formal data acquisition (Burgman, 2016; Hemming et al., 2018a). In these conditions, expert judgements are considered as a source of data (Cooke, 1991). In a structured process, experts are assisted to convert their knowledge into quantitative estimates (Hemming et al., 2018b). In the study of salmon diseases, subjective probability-estimation protocols have been applied to inform decision making in the control of infectious salmon anemia (ISA), calculating a likelihood ratio for risk factors (Gustafson et al., 2005, 2014).

In this study we used a structured expert elicitation process to identify and prioritize risk and protective factors associated with severe outbreaks of SRS. Our approach is especially useful to inform decisions with sparse information available, systematically reducing individual and group heuristic biases (Burgman, 2016; Hemming et al., 2018a; Kuhnert et al., 2010). Thereby providing insights to inform prevention and control strategies for the most important infectious disease affecting farmed salmon in Chile.

2. Materials and methods

The Delphi method, developed for the United States Air Force in 1960, is a common protocol to integrate experts' judgements who analyze a given problem (Dalkey and Helmer, 1963). In the Delphi

process, experts make successive individual estimates, receiving anonymous feedback until a pre-specified level of agreement is achieved (Linstone and Turoff, 1975; McBride et al., 2012). Thus, the Delphi method promotes consensus among a group of experts in iterative steps, and it has advantages compared with an unstructured elicitation process (Kerr and Tindale, 2004).

The IDEA approach (Investigate, Discuss, Estimate, Aggregate), an adaptation of the Delphi method, proposes a structured protocol to collect and aggregate expert judgements of uncertain parameters (Hanea et al., 2017). In the IDEA protocol, experts provide individual estimates in subsequent rounds of elicitations. Additionally, IDEA incorporates an open discussion phase to interchange arguments among experts (Hanea et al., 2016, 2017; Hemming et al., 2018a, 2018b). IDEA does not encourage consensus (Hanea et al., 2017). Indeed, to improve accuracy in estimates, experts need to openly discuss arguments and rationales of feedbacks, maintaining anonymity in the individual estimates (Bolger et al., 2011; Rowe et al., 2005). The IDEA method improves the accuracy and reliability of experts' judgments, focusing on carefully constructed uncertainty bounds for coping with overconfidence (Burgman, 2016).

Fig. 1 presents a schematic representation of the protocol applied for the expert elicitation process, based on IDEA approach (Hanea et al., 2017; Hemming et al., 2018b). We divided the process in three stages: 1) Pre-elicitation, 2) Elicitation and 3) Post-elicitation.

2.1. Stage 1: pre-elicitation

2.1.1. Recruit experts

We determined two criteria to identify experts. First, we sought individuals with experience in SRS sanitary management or/and SRS epidemiology research. Second, we balanced the number of experts according to sectors, including government services, industry, academia and laboratories. Most of the identified experts were Chilean veterinarians with experience in diagnosis, treatment or epidemiological studies of SRS. A total of 16 experts were invited to participate in the study, three declined the invitation. As a result, we selected a group of 13 experts with gained experience in health management of Chilean farmed salmon, representing government services (23.1%), industry (53.8%) and academia/laboratories (23.1%).

2.1.2. Identify risk and protective factors

First, based on literature review and preliminary interviews with experts, we identified 45 factors that were reported as risk or protective factors for infectious diseases affecting farmed salmon. Subsequently, via email, each expert received the list of factors with their corresponding definitions. At this stage, there was no contact among participants. Experts were asked to select a maximum of 20 factors considered as the most relevant, according their knowledge and experience. Here, experts had the opportunity to include any other risk or protective factor for further discussion. From the initial list of 45 factors, six were not selected by any expert, so these were discarded. On the other hand, experts provided ten new factors that were included in a second list of 49 factors.

The revised list of 49 factors was circulated among participants. In the revised list, the factors were ordered by frequency (number of times the factor was selected by an expert). Participants selected again a maximum of 20 factors considered as the most relevant for them. As a result, we excluded 26 factors that were selected only by three or fewer experts. In addition, five pairs of factors were merged into five factors because of similarities. Finally, a total of 11 risk factors and 7 protective factors were pre-selected for the next stage.

2.1.3. Establish the event of interest

In Chile, nearly 90% of Atlantic salmon farms reported SRS mortalities at some point during the production cycle between 2010 and 2014 (Mardones, 2016). Consequently, we emphasized the severity of

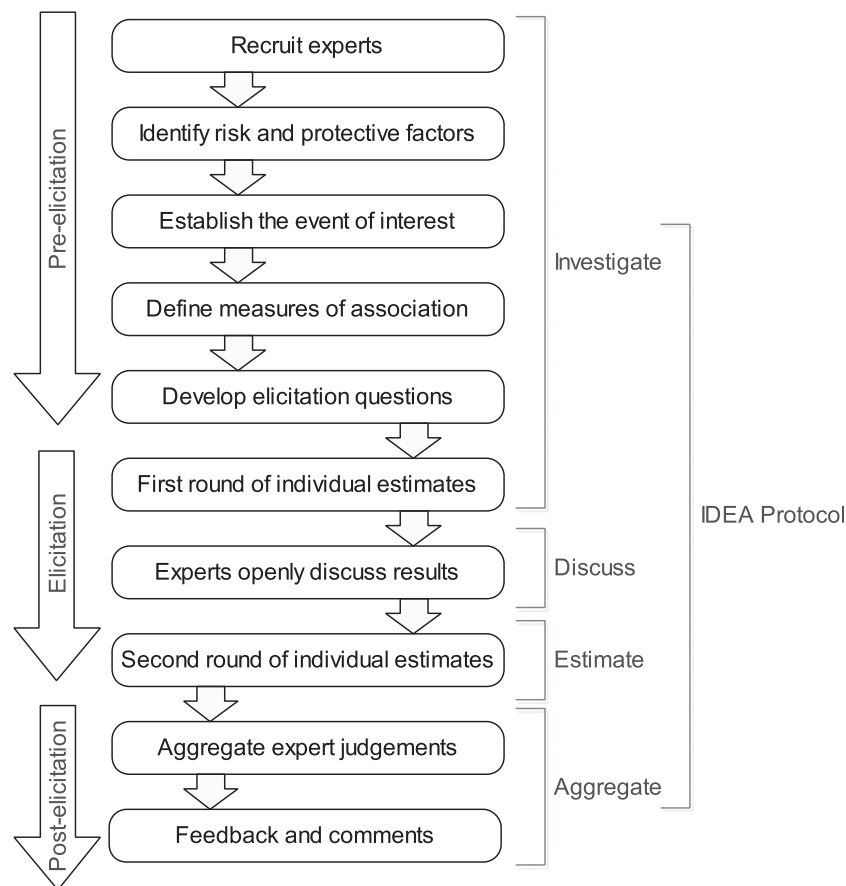


Fig. 1. Flowchart of the four stages for the expert elicitation process.

SRS outbreaks in Atlantic salmon farms rather than focusing on its occurrence.

We defined as the event of interest two opposed but realistic scenarios: a farm experiencing severe outbreaks of SRS, and a farm that had SRS but with low mortalities throughout the production cycle. To facilitate the expert elicitation process, we established that a “severe outbreak” took place if an Atlantic salmon farm would present an early onset of SRS followed by a steep epidemic curve of SRS-specific mortalities, around three-fold that expected, after 6-month fish were transferred to salt water (Fig. 2). In the “low mortalities” scenario, the Atlantic salmon farm would present average SRS-specific mortalities, without a steep epidemic curve around 6-month fish were transferred to salt water (Fig. 2).

2.1.4. Define measures of associations

The concept of probability has a dual nature (O'Hagan, 2006). In the standard statistical approach, probability is normally understood as the relative frequency at which an event is expected to occur (Burgman, 2005). In the other side, associated with Bayesian statistics, when the frequency of an event is unknown, probability is understood as the degree of belief, based on evidence and experience, about the frequency of an event (O'Hagan, 2006).

We considered two measures of associations to explore the relationship between the perceptions of a given risk or protective factor and the event of interest: Relative Risk (RR) and Absolute Risk Reduction/Increase (ARR/ARI) (Dohoo et al., 2007). In this study, we calculated risk estimates based on expert judgments of probabilities. Thus, risk measures were not assessed for statistical significance. Table 1 illustrates how to obtain RR and ARR/ARI based on the two aforementioned scenarios of SRS (severe outbreaks and low mortalities). The contingency table indicates the frequency of farms, out of

a total of ten, with *severe outbreaks* of SRS, in which a specific factor is present (a), and in how many a specific factor is absent (c). In addition, the contingency table indicates in how many of ten farms with *low mortalities* of SRS the factor is present (b), and in how many the factor is absent (d).

2.1.4.1. Relative Risk. Relative Risk (RR) compares the probability of developing a severe outbreak of SRS when the factor under study is present, with the probability of developing a severe outbreak of SRS when the factor under study is absent. RR has values between 0 and infinity. If $RR > 1$, the presence of the factor increases the probability that the farm would experience a severe outbreak of SRS; in these cases, the factor is referred to as a risk factor. If $RR < 1$, the factor decreases the probability that a farm would experience a severe outbreak of SRS; here, we referred such factor as a protective factor. If $RR = 1$, the presence of the factor has no association to the severity of SRS. From Table 1, RR is estimated as $(a/(a + b))/(c/(c + d))$.

2.1.4.2. Absolute Risk Reduction/Increase. In some situations, relative measures of risk could amplify the effects of associations (Dohoo et al., 2007). Absolute Risk Reduction/Increase (ARR/ARI) estimates, in absolute terms, the percentage change of the probability that a severe outbreak would happen under the presence of a risk or protective factor. From Table 1 ARR/ARI is estimated as $(a/(a + b)) - (c/(c + d))$.

2.1.5. Develop elicitation questions

The IDEA protocol elicits quantitative judgements, providing a bound of uncertainty around the best estimate (Hanea et al., 2017). Establishing uncertainty, or a bound between the realistic smaller values and greater values, it prompts experts to consider different possibilities before making their best estimates (Morgan and Henrion, 1990;

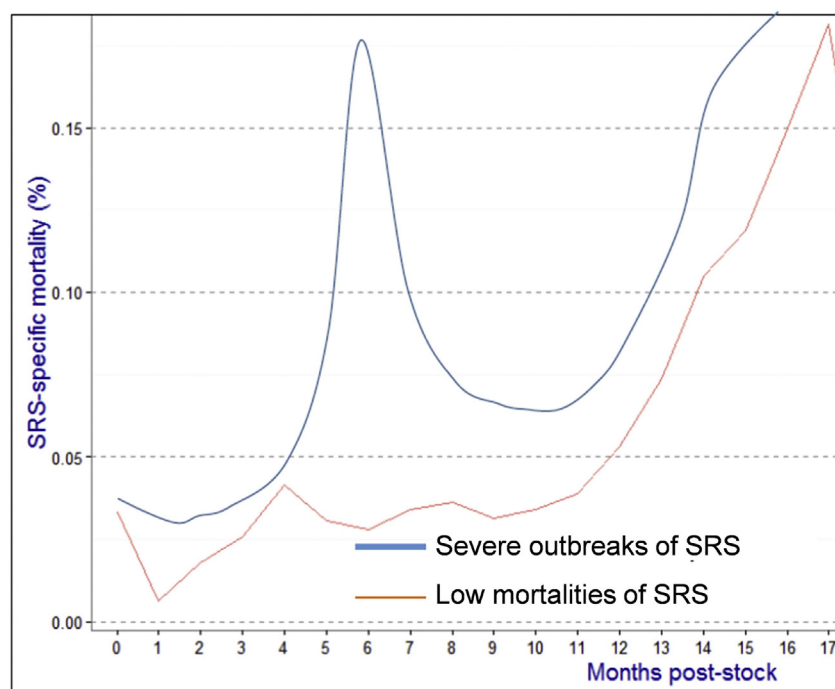


Fig. 2. A farm with a severe outbreak of SRS would present a steep epidemic curve of SRS-specific mortalities, around three-fold that expected, after 6-month fishes were transferred to salt water. A farm with low mortalities caused by SRS would present average SRS-specific mortalities for a productive cycle. Representation of the epidemic curves and mortality levels were obtained from official data provided by SERNAPESCA.

Speirs-Bridge et al., 2010). In the three-step format, questions focus on eliciting probabilities (Burgman, 2016). In this format, prior to elicit their best estimate, experts provide their upper and lower bounds, in terms of the lowest probability and the highest probability that an event will occur (Hemming et al., 2018b). Finally, experts indicate their best estimate about the frequency of an event occurrence (Hanea et al., 2017).

In this study, we used the three-point format question to elicit experts' estimates. However, we re-framed probability questions as natural frequencies to facilitate elicitation of expert judgments (Gigerenzer and Edwards, 2003). We formulated each question as a number of farms over ten in which the factor is present (see an example in Table 2). Experts were asked to indicate the plausible minimum value (step 1), the plausible maximum value (step 2) and the best estimate (step 3) about the presence of a factor (risk or protective) in each of both scenarios for SRS (severe outbreaks and low mortalities) described in section 2.1.3.

Risk and protective factors were treated as dichotomous variables, indicating their presence or absence in both scenarios: severe outbreaks of SRS and low mortalities caused by SRS. For 13 factors, dichotomies were obtained directly based in the literature review. The final definitions for factors' dichotomies were discussed and agreed with experts. To exemplify dichotomies, Table 2 presents the questions for the risk factor fouling of cage (see definitions in Table 3).

For four factors, specific cut-off levels were assessed by the experts: the number of anti-parasite bath treatments before the onset of SRS in a farm (anti-parasitic bath-treatments), the number of active farms in a ring of 5 km (density of farms in neighborhoods), the number of stocked fish in a farm (stocked fishes) and the critical distance (in km) to a salmon farm experiencing SRS (closeness to a farm with SRS). For these four factors, previously to answer the probability questions, experts

provided quantitative estimates to determine cut-off values, including the minimum plausible value, the maximum plausible value and the best estimate for both scenarios (severe outbreaks of SRS and low mortalities caused by SRS). Cut-off values were calculated by the Receiver Operating Characteristics (ROC) analysis, using the package “pROC” in the software R (Fawcett, 2006; R Development Core Team, 2018; Robin et al., 2017).

2.2. Stage 2: elicitation

2.2.1. First round of individual estimates

We organized an introductory meeting with each expert, explaining motivations, expectations and context for the expert elicitation process. In the same meeting, the first author presented the methodological approach, clarifying doubts about the protocol and questions. Each expert reviewed the list of factors and both scenarios for SRS (severe outbreaks and low mortalities), reducing the language uncertainties among participants (Burgman, 2005).

A week later, we sent an email to the experts containing the questions for 18 factors in a spreadsheet and instructions to complete them. For each factor, experts made independent assessments for both scenarios. We encouraged experts to circumscribe their responses to a sample of farms where they have worked or visited, which greatly exceeds 10 farms with severe outbreaks of SRS and 10 farms with low mortalities caused by SRS. If an expert does not have direct experience about the presence or absence of a particular factor, we encouraged them to consider their knowledge in salmon diseases or omit that factor.

2.2.2. Experts openly discuss results

Prior to the discussion stage, we aggregated individual estimates of the first round, generating graphs and measures of central tendency for

Table 1

Contingency table of 2×2 to calculate Relative Risk (RR) = $(a/(a + b))/(c/(c + d))$, and Absolute Risk Reduction/Increase (ARR/ARI) = $(a/(a + b)) - (c/(c + d))$.

	Ten farms with severe outbreak of SRS	Ten farms with low mortality of SRS	Total
Number of farms with the PRESENCE of factor	A	b	a + b
Number of farms with the ABSENCE of factor	C	d	c + d
Total	a + c = 10	b + d = 10	

Table 2Question used to elicit experts estimates for the risk factor *Fouling of cage*. Original texts were in Spanish.

Fouling in cage	Lower plausible number	Greater plausible number	Best estimate
Consider 10 farms that have suffered one or more severe outbreaks of SRS during the last two productive cycles.			
Of these 10 farms, how many had lack of maintenance and/or inadequate replacement of fishnets?			
Now consider 10 farms that have suffered only low mortalities in the last two productive cycles. Of these 10 farms, how many had lack of maintenance and/or inadequate replacement of fishnets?			

each question (see details in 2.3.1. Aggregate experts' judgements). Then, we organized a workshop where experts accessed the results of the first round and openly discussed them (maintaining the anonymity of individual estimates). This workshop was facilitated by the first author of this manuscript. For each factor, the facilitator presented results from first round, stimulating discussion particularly in questions with higher inter-expert dispersion. In this stage, the risk factor Density of cage was discarded as it overlaid with the factor Number of stocked fishes in farm. It was not clear how the impact of Density of cage can vary along the production cycle, generating confusion among experts.

2.2.3. Second round of individual estimates

Immediately following the group discussion, experts were asked to conduct a second round of anonymous and independent estimates for each factor. A week later, we sent a spreadsheet with questions for the four factors that were re-estimated as continuous variables, now converted to dichotomous variables through identification of the cut-off value based on an ROC analysis.

2.3. Stage 3: post-elicitation

2.3.1. Aggregate experts' judgements

Individual estimates were simply aggregated by their means (Hemming et al., 2018a). The coefficient of variation (CV) was used as a measure of the inter-expert variation of the individual estimates (Adams-Hosking et al., 2016). Following the second round, the estimates were used to calculate the RR and the ARR/ARI.

2.3.2. Feedbacks and comments

Results were circulated among participants via email, receiving feedbacks and comments for the final report. Feedbacks were also captured via video conference.

Table 3

Relative Risks (RR) and Absolute Risk Reduction/Increase (ARR/ARI) of risk and protective factors for SRS. CdV = coefficients of variation.

ID	Factor	Description	Type	RR	CdV	ARI/ARR	CdV
Risk factors							
1	Anti-parasitic bath-treatments	Increased frequency of anti-parasitic bath-treatments before the onset of clinical SRS	Stressor	3.34	62.9%	42.30%	46.1%
2	Fouling of cage	Failure to maintain and/or adequately exchange fish nets	Stressor	3.24	44.7%	49.50%	22.3%
3	Density of farms in neighborhoods	Number of active farms in a ring of 5 km	Source	2.77	36.8%	45.96%	27.9%
4	Presence of sea lions	Attacks of sea lions due to bad condition of the sea lions meshes	Stressor	2.53	41.9%	37.79%	37.5%
5	Closeness to a farm with SRS	Distance (km) to a farm while undergoing a severe outbreak of SRS	Source	2.44	29.1%	39.86%	31.2%
6	Stock season	Season in which the stocking takes place (summer, autumn, winter, spring)	Stressor	2.41	31.9%	37.94%	31.7%
7	Presence of caligus	High re-infestation of caligus	Stressor	2.30	21.7%	35.93%	28.1%
8	Stocked fishes	Number of stocked fish in a farm	Stressor	2.22	23.4%	35.83%	26.3%
9	Early mature salmonids	Early presence of mature salmonids (below 1.5 kg.)	Source	1.91	46.1%	27.33%	60.8%
10	Previous SRS in farm	Previous presence of recurrent severe outbreaks of SRS in a farm	Source	1.71	27.9%	24.29%	58.6%
Protective factors							
11	Synchronized farm rest period	Synchronized farms rest period in a neighborhood	Protector	0.42	30.9%	42.23%	26.9%
12	Opportune diagnosis	Early clinical-laboratory diagnosis of SRS	Protector	0.47	28.8%	39.87%	33.4%
13	Necropsy training	Personnel properly identify injuries associated with pathologies/sampling	Protector	0.50	48.0%	37.47%	53.1%
14	Current and oxygen	Current and oxygen conditions in farms	Protector	0.51	39.2%	35.03%	51.7%
15	Productive elimination	Daily elimination of misfits and stragglers	Protector	0.55	38.2%	31.48%	54.3%
16	Eliminate fish mortality	High biosecurity conditions in the elimination of mortalities	Protector	0.56	44.6%	31.61%	67.3%
17	Proper use of vaccines	Recommendations from vaccine providers are met with respect to the Grade Days (UTA)	Protector	0.78	24.4%	15.11%	91.7%

3. Results

3.1. Risk and protective factors

From the screened list of 49 factors, a set of ten (20.4%) were considered as risk factors ($1.7 < RR < 3.4$), and seven (14.3%) as protective factors ($0.4 < RR < 0.8$) (Table 3). Anti-parasitic bath-treatments, fouling of cage and density of farms in neighborhoods appeared as the riskiest factors ($ARR/ARI > 40\%$) (Table 3, Fig. 3). On the other hand, synchronized farm rest period, opportune diagnosis and necropsy training appeared as the most important protective factor ($ARR/ARI > 37\%$) (Table 3, Fig. 3). The results of ROC analysis are included in the Appendix.

3.2. Expert performance through the process

Between the first and second round, the CV of participants' best estimates were systematically reduced (from an average of 45.9% in the first round to 25.3% in the second round). To exemplify the effect of the workshop reducing dispersion among experts' estimates, we present results for the factor presence of sea lions. In the first round (Fig. 4A and B), a relatively high CV was observed for Low mortalities (45.3%) and Severe outbreak (34.9%). In the second round (Figs. 4C y 4D), after the group discussion, the CV were reduced to 25.8% and 16.5% respectively. Interestingly, the best estimate for this factor remained stable between rounds, but inter-expert variation was reduced.

4. Discussion

In this study, we used an expert-based approach to systematically elicit data from a group of health management experts in salmon aquaculture. We identified ten risk factors and seven protective factors that appear to be related with severe outbreaks of SRS. For the 17 factors, we calculated their RR and ARR/ARI as risk measures of

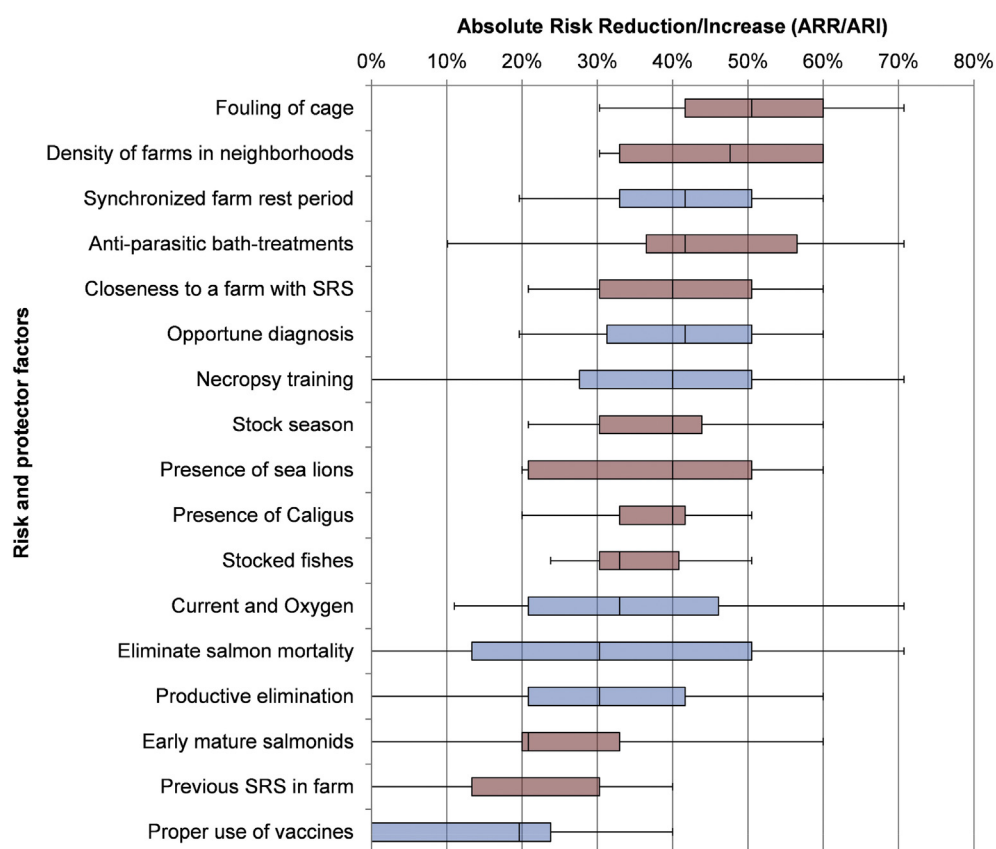


Fig. 3. Box plots for ARR/ARI for risk (red) and protective (blue) factors obtained from the expert elicitation process. The median is represented by the line that divides the box into two parts. The box contains the 50% of experts' estimates. The lines in each extreme of the box represent the 25% of experts' estimates. Variables are sorted from high to low ARR/ARI. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

associations for a scenario of severe outbreaks of SRS in a conventional production cycle for farmed Atlantic salmon. The expert elicitation process included two rounds of individual estimates, and a workshop between rounds. In the workshop, experts were able to present arguments and rationalities for their judgments. As a workshop result, inter-expert variations of the individual estimates were systematically reduced. We showed that a structured and rigorous process for expert elicitation could provide synthesized information to better understand and prioritize research aimed at avoiding severe outbreaks of SRS.

Our results are consistent with preliminary studies that have identified risk factors associated to severe outbreak of SRS (e.g. Branson and Nieto Díaz-Muñoz, 1991; Cusack et al., 2002; Larenas et al., 1995; Rees et al., 2014). However, our study has also identified novel risk and protective factors that have not been investigated, either because they are not part of monitoring programs, or simply because there is no quantitative data to directly record them.

Novel factors that appear to increase the risk of severe outbreaks of SRS found in this study were fouling of cages and presence of sea lions. Currently, in Chile there is no monitoring data available or objective parameters to determine risk thresholds or impact categories for both factors. Fouling of cage may exert its effect by increasing the stress in fish by reducing pen ventilation and oxygen availability. To address fouling, it is important to begin identifying objective and measurable factors of cleanliness of fishnets, anti-fouling systems, environmental conditions of the concession and the effect of the time of year. Similarly, presence of sea lions may exert its influence by increasing the stress in fish. In this case, by the predatory behavior of these pinnipeds who constantly attack fish through the nets. It is also necessary to determine parameters to evaluate the impact of presence of sea lions, for example evaluating variables such as mortality by sea lions attack, number of attack events by sea lions in the cage or number of events in which the sea lions enter the module, but not the cage.

Novel factors that could reduce the risk of severe outbreaks of SRS

were identified through the expert elicitation process (i.e., synchronized farm rest period, opportune diagnosis and necropsy training). Synchronized farm rest period appeared as the most important protective factor to reduce the risk of severe outbreaks of SRS. Following between production cycles is a frequent practice in Chile, aiming to interrupt the disease cycle by removing the host for a period at which pathogen cannot survive. A recent study by Price et al. (2017) has provided the first evidence that following by 3 months reduces the risk of SRS outbreaks in Atlantic salmon farms. Oxygen availability and currents are two environmental factors closely related (current and oxygen). Farms located in areas with low current speed will probably have events of low oxygen, while farms located in exposed areas with high speed currents oxygen would not be a limiting factor. As mentioned above, oxygen availability is a factor that conditions fish response to challenging pathogens such as *P. salmonis*.

We organized risk and protective factors in a conceptual model (Fig. 5). In this model, risk factors are classified as stressors or sources. Stressors are risk factors that could drive the magnitude of the disease outbreaks (anti-parasitic bath-treatments, fouling of cage, presence of sea lions, stock season, presence of caligus, early maturation of salmonids and stocked fishes). Early mature fish may be more at risk of becoming infected because of the stress of sexual maturation (Good and Davidson, 2016). On the other side, sources are risk factors associated with exposure to the pathogen (density of farms in neighborhoods, closeness to a farm with SRS and previous SRS in farm).

We further classified protective factors according to levels of intervention: primary, secondary or structural (Fig. 5). Structural interventions focus primary in control risk factors categorized as a source of the pathogen, including synchronized farm rest period, farms located in areas with appropriate current and oxygen, and proper use of vaccines. The secondary interventions focus on increasing professional capacities for early detection of diseases. In this category, we included opportune diagnosis and necropsy training. Finally, primary interventions include

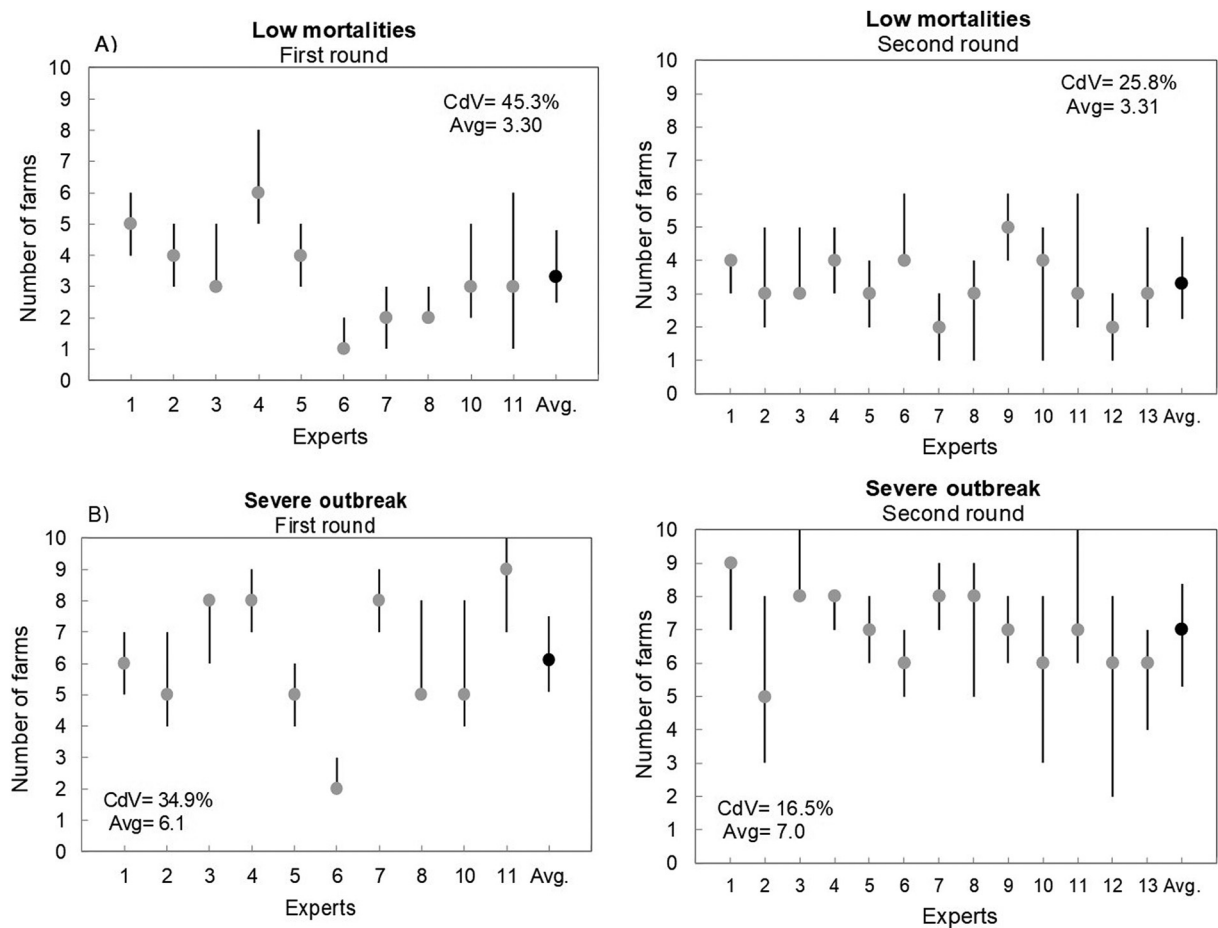


Fig. 4. Experts' first and second round estimates for the presence of sea lions in A) low mortalities caused by SRS, and B) severe outbreaks of SRS. Uncertainty bars represents the minimum and maximum estimate for each expert. CdV = Coefficients of variation. Avg = mean.

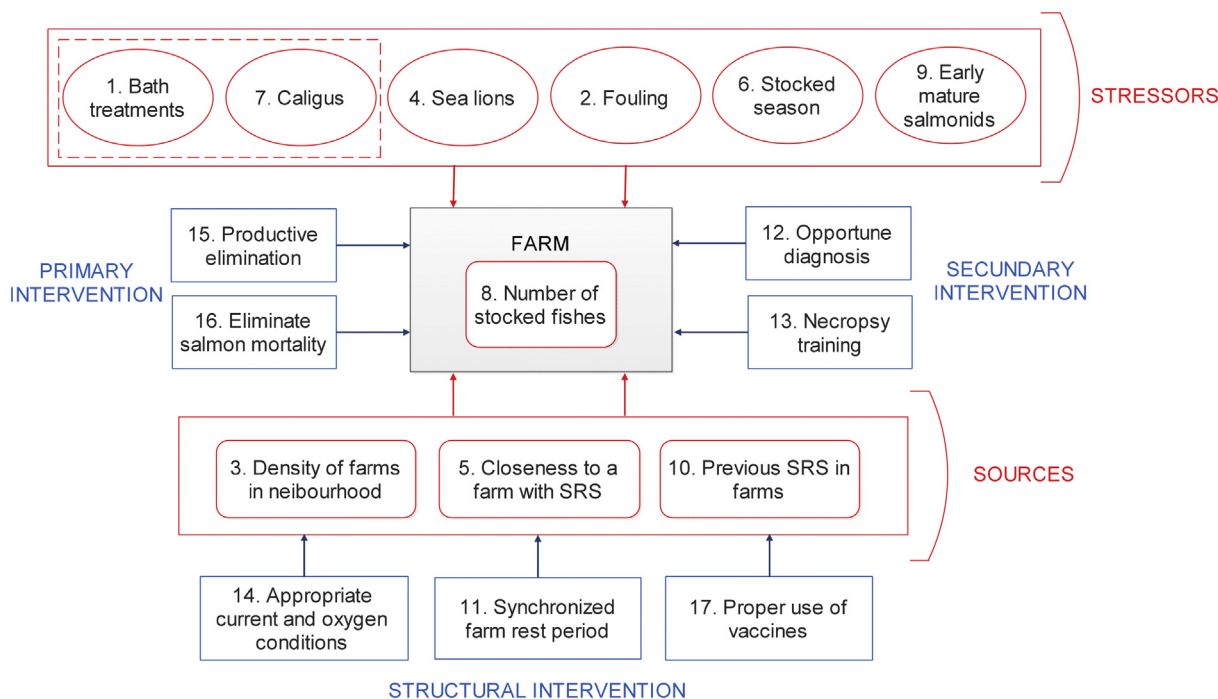


Fig. 5. The intervention model for preventing and controlling severe outbreaks of SRS.

protective action in the farms, included productive elimination of fishes with unfavorable conditions for normal development, and eliminating systematically salmon mortality.

Expert elicitation processes may be subject to errors and bias associated with the qualitative based approach (O'Hagan, 2006). We confronted these during the expert elicitation process. One difficulty was the systematic uncertainty of language associated to risk and protective factors. Particularly with factors that were not subject to natural measurements, due to the absence of data or because of their qualitative nature. Similarly, experts did not differentiate between the two scenarios based on threshold mortality levels. As a differentiating criterion, we established the presence of a steep epidemic curve of SRS-specific mortalities around six-months fishes were transferred to salt water established criterion. This last methodological decision facilitated the expert elicitation process; however, it may lead to confusion if concepts were not properly defined. For this, an iterative work was carried out with the experts to adjust, correct or guide the interpretations of the factors' definitions towards a common understanding. During the workshop, the experts were able to dialogue and agreed on definitions.

A second challenge was the potential difficulties of experts to estimate probabilities based on subjective judgments (Martin et al., 2012). Additionally, individuals cannot cognitively process multiple interacting and confounding factors on an outcome. To deal with these biases, prior to the first round of estimates, individual meetings were held with each expert to explain the three-step format question, scenarios and definitions and the rationalities behind the qualitative-based approach.

A relevant limitation refers to the interpretation of the risk assessment. In this study, RR were calculated from univariate assessments of the risk/protective factors, and results were not adjusted for distortion or corrected for confounding effects from the other predictors. Therefore, we cannot establish statistical inferences from results. Despite the difficulties and potential biases in the estimates of probabilities by expert judgments, the absence of data or evidence does not limit the need to make decisions for the management of highly harmful

outbreaks of SRS. Indeed, decision makers require to systematically evaluate the relative risk of factors despite the absence of data or evidence (McCarthy et al., 2004). Consequently, evidence based-management and expert based knowledge are important and complementary sources of data disease management and control (Fazey et al., 2006).

In Chile, the salmon industry has faced a myriad of challenging issues including production stability, sustainability (e.g., environmental, sanitary, economic, etc.), and social transformations (Barton and Román, 2016; Peralta et al., 2015). Currently, SRS is one of the main obstacles for the salmon industry development and transformation towards becoming less antibiotic dependent. In this study, we identified ten risk factors and seven protective factors that could prevent and control severe outbreaks of SRS. We used a qualitative based-expert approach to calculate risk estimator measures. For each factor, we calculated the RR and ARR/ARI. We identified novel factors that may increase the risk of severe outbreaks of SRS, such as fouling of cage, density of farms in neighborhoods and presence of sea lions; we also found novel factors that may reduce the risk of severe outbreaks of SRS, such as synchronized farm rest period, opportune diagnosis and necropsy training. Results are important to shift the attention placed on common aspects of salmon disease management, towards new critical issues which have been commonly overlooked. A structured and rigorous expert elicitation process is effectively providing scientific information to better prevent and control diseases in salmon farms.

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Appendix A. Appendix

Table A1
Results of ROCs calculations for four risk factors.

Factor	Description	Unit	Area under the curve	Cut-of value
Anti-parasitic bath-treatments	Increased frequency of anti-parasitic bath-treatments before the onset of clinical SRS	Number of bath-treatments	0.858	2
Density of farms in neighborhoods	Number of active farms in a ring of 5 km	Number of active farms	0.822	2
Stocked fishes	Number of stocked fish in a farm	Number of fishes	0.925	950.000
Closeness to a farm with SRS	Distance (km) to a farm while undergoing a severe outbreak of SRS	Kilometers	0.878	6

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